

7. Criminals will try to hide their activities but forensic scientists, using biological and chemical testing, are able to find evidence that reveals what happened.

(a) DNA profiling is commonly used to identify criminals.

(i) State where DNA is found in the cell.

[1]

.....

(ii) There were three suspects in a crime. Suspect 1 did not have an alibi but suspects 2 and 3 did. Initially it was thought that suspect 1 was the criminal. DNA samples were collected from the crime scene.



Compare the DNA samples above to explain whether having an alibi or not is sufficient to decide on guilt.

[2]

.....

.....

.....

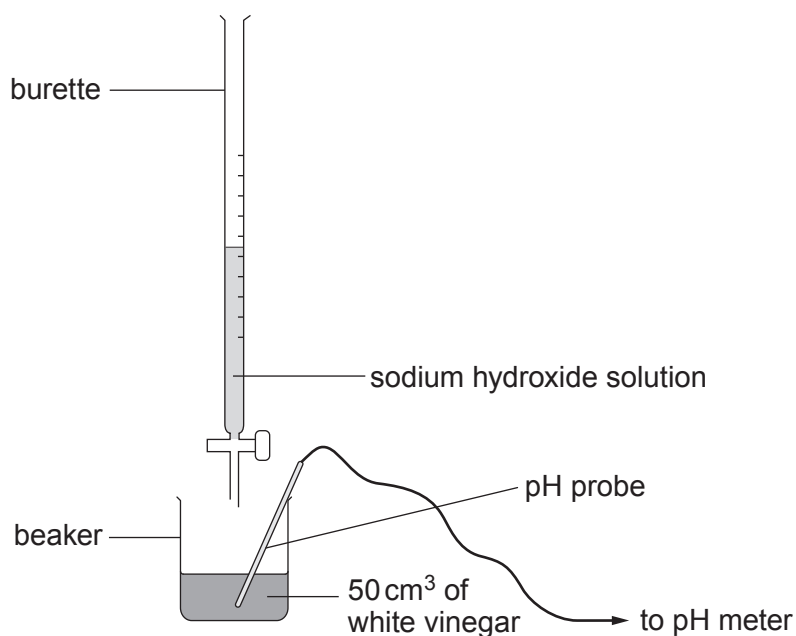
(iii) State **one other** use of DNA profiling.

[1]

.....



- (b) It is suspected that supplies of vinegar have been tampered with by adding a strong acid. White vinegar usually contains ethanoic acid at a concentration of 0.01 mol/dm^3 . Samples of the vinegar are taken and tested by titrating against 0.5 mol/dm^3 sodium hydroxide solution using the apparatus below.



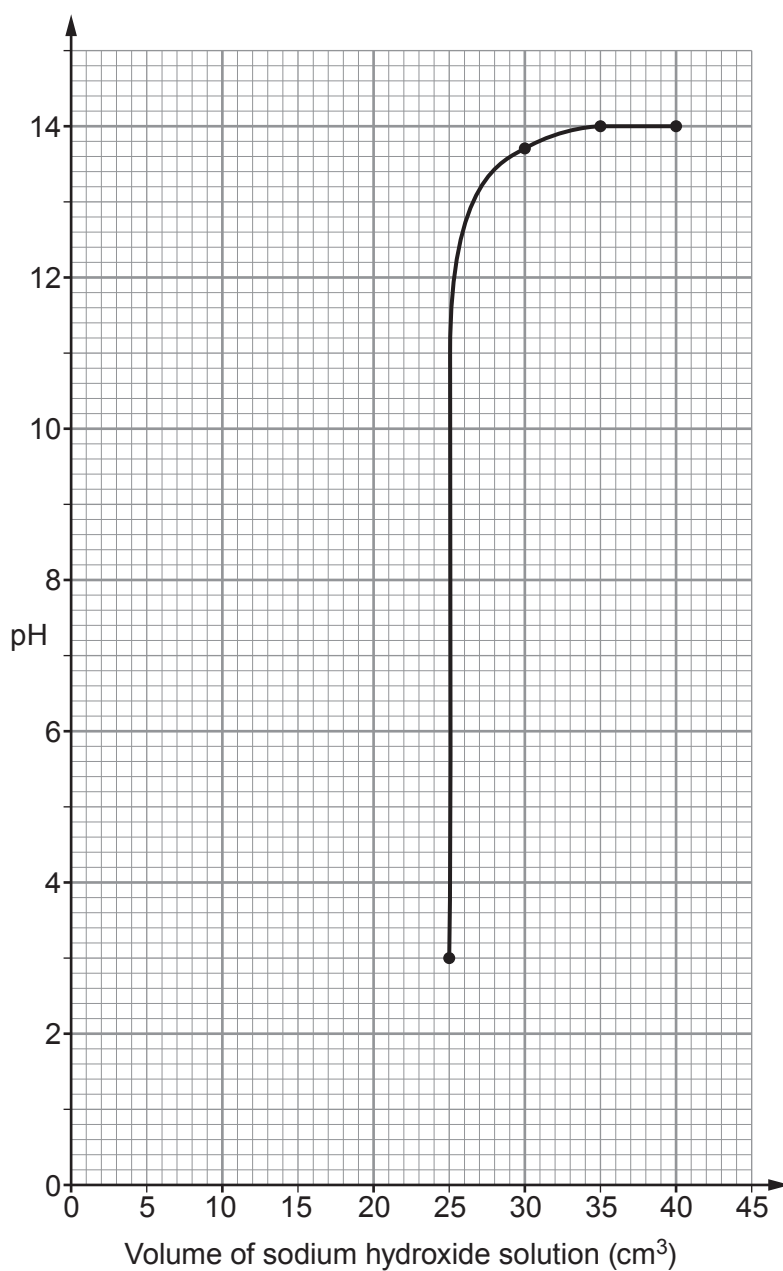
A pH meter is used to monitor the pH of the solution in the flask. The results are shown in the table below.

Volume of sodium hydroxide solution (cm^3)	pH reading
0	1.5
5	1.6
10	1.7
15	1.9
20	2.2
25	3.0
30	13.7
35	14.0
40	14.0



- (i) Use the data in the table to complete the graph on the grid below.

[3]



- (ii) Use the graph to find the volume of sodium hydroxide solution required to neutralise the white vinegar.

[1]

volume = cm³



- (iii) Use the information on pages 16 and 17 and the equation

$$\text{concentration of white vinegar} = \frac{\text{concentration of sodium hydroxide} \times \text{volume of sodium hydroxide}}{\text{volume of white vinegar}}$$

to determine whether or not the samples had been tampered with. [3]

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- (iv) Complete the word equation for the reaction between an acid and a base. [2]

acid + base →

- (v) Explain why taking multiple readings will increase the accuracy of the result. [2]

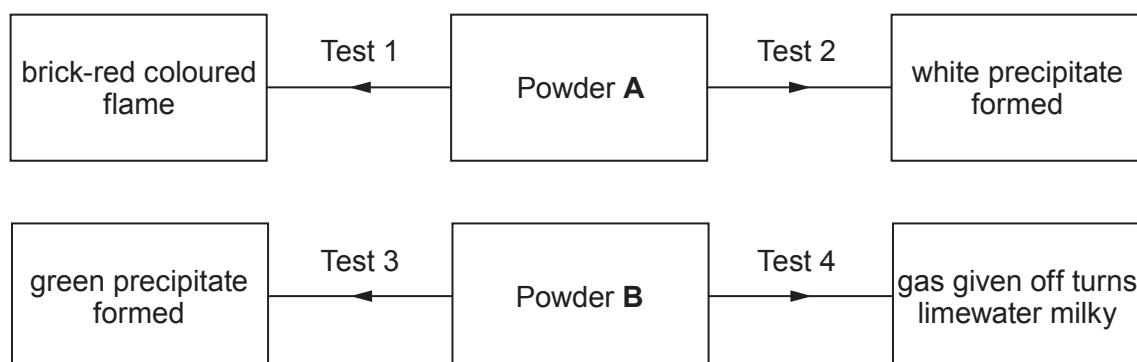
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- (c) Powders **A** and **B** are collected at a crime scene. The following tests are used to identify them.

Test	Action
1	Flame test
2	Add silver nitrate solution
3	Add sodium hydroxide solution
4	Add dilute hydrochloric acid and bubble gas through limewater

The results are given below.



Identify powders **A** and **B**.

[4]

Powder **A**:

Powder **B**:

END OF PAPER





THE PERIODIC TABLE

Group

1

2

3

4

5

6

7

0

1 H 1 Hydrogen																		4 He 2 Helium	
																		19	20
																		F	Ne
																		Fluorine	Neon
																		9	10
																		35.5	40
																		Cl	Ar
																		Chlorine	Argon
																		17	18
11	12	14	16															80	84
B	C	N	O															Br	Kr
Boron	Carbon	Nitrogen	Oxygen															Bromine	Krypton
5	6	7	8															35	36
27	28	31	32															35.5	40
Al	Si	P	S															Cl	Ar
Aluminium	Silicon	Phosphorus	Sulfur															Chlorine	Argon
13	14	15	16															17	18
70	73	75	79	65	59	59	56	55	52	51	48	45	44	43	42	41	40	38	36
Ga	Ge	As	Se	Zn	Ni	Co	Fe	Mn	Cr	V	Ti	Sc	Y	Tc	Mo	Nb	Zr	Sr	Kr
Gallium	Germanium	Arsenic	Selenium	Zinc	Nickel	Cobalt	Iron	Manganese	Chromium	Vanadium	Titanium	Scandium	Yttrium	Technetium	Molybdenum	Niobium	Zirconium	Strontium	Krypton
31	32	33	34	30	28	27	26	25	24	23	22	21	39	43	42	41	40	38	36
115	119	122	128	112	106	103	101	99	96	93	91	89	88	86	84	83	82	80	78
In	Sn	Sb	Te	Cd	Pd	Rh	Ru	Tc	Mo	Nb	Zr	La	Ba	Rb	Xe	I	Xe	Xenon	Xenon
Indium	Tin	Antimony	Tellurium	Cadmium	Palladium	Rhodium	Ruthenium	Technetium	Molybdenum	Niobium	Zirconium	Lanthanum	Barium	Rubidium	131	127	131	Xenon	Xenon
49	50	51	52	48	46	45	44	43	42	41	40	39	56	37	54	53	54	Xenon	Xenon
204	207	209	210	201	197	192	190	186	184	181	179	139	226	223	222	210	222	Rn	Rn
Tl	Pb	Bi	Po	Hg	Au	Pt	Os	Re	W	Ta	Hf	Ac	Ra	Fr	86	85	86	Radon	Radon
Thallium	Lead	Bismuth	Polonium	Mercury	Gold	Platinum	Osmium	Rhenium	Tungsten	Tantalum	Hafnium	Actinium	Radium	Francium	86	85	86	Radon	Radon
81	82	83	84	80	79	78	76	75	74	73	72	89	88	87	86	85	86	Radon	Radon

Key

Ar

Symbol

Name

Z

relative atomic mass

atomic number